

Farmer GPT – AI Assistant for Agriculture

Antler Pitch Deck

1. Problem

Millions of farmers in India still struggle to access reliable and timely agricultural information.

Current challenges:

- Information is fragmented across YouTube videos, local dealers, WhatsApp groups, and government portals.
- Most existing solutions are not personalized or available in regional languages.
- Farmers often receive inaccurate advice, leading to crop loss, unnecessary spending, and lower productivity.
- AI chatbots today can hallucinate and are not reliable enough for agriculture-specific decision making.

2. Solution

Farmer GPT is an AI-powered agricultural assistant designed to provide farmers with accurate, context-aware, and accessible farming guidance.

The platform helps users with:

- Crop recommendations
- Fertilizer guidance
- Weather-based suggestions
- Pest and disease information
- Government schemes and subsidies
- Localized agricultural support

Built using Large Language Models (LLMs) integrated with Retrieval-Augmented Generation (RAG) to reduce hallucinations and improve reliability.

3. Why Now?

- Smartphone and internet penetration in rural India is increasing rapidly.
- Farmers are already consuming agricultural content digitally.
- India is seeing rapid adoption of AI tools across sectors.
- Government focus on AgriTech and Digital Public Infrastructure creates a strong ecosystem for scale.

AI can now become a practical everyday assistant for rural users.

4. Market Opportunity

- India has 140M+ farmers.
- Agriculture contributes significantly to India's economy.
- AgriTech adoption is still in early stages, leaving a massive untapped market.
- Opportunity to expand into:
 - Financial products
 - Crop insurance
 - Marketplace integrations
 - Supply chain insights
 - Rural commerce

Farmer GPT can become a trusted digital layer between farmers and agricultural services.

5. Product & Technology

Core stack:

- LLM-powered conversational AI
- Retrieval-Augmented Generation (RAG)
- Structured agricultural knowledge base
- Multilingual support

- Future voice-based interaction system

Key focus:

- Reliable responses
- Context-aware retrieval
- Low hallucination rates
- Accessibility for non-technical users

6. Traction / Progress

- Initial product architecture under development.
- Working on retrieval pipelines and agricultural knowledge integration.
- Researching multilingual workflows and rural accessibility.
- Background in building real-world products and managing operational systems.

7. Founder

Rajamuri Sai Rithik Reddy

- B.Tech Computer Science & Data Science student
- College President at CR Rao AIMSCS
- Experience leading teams, managing large-scale events, and operational execution
- Built software products including AI systems and hospitality management platforms
- Strong interest in AI, startups, and building scalable products for India

8. Vision

To build the most trusted AI assistant for farmers in India by making agricultural intelligence accessible, reliable, multilingual, and easy to use.

Long-term vision: Create an AI-powered rural operating system connecting farmers with information, financial services, and digital infrastructure.

